Section 4.10 : Table Of Laplace Transforms

Table of Laplace Transforms

	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1.	1	$\frac{1}{s}$
2.	$\mathrm{e}^{a\,t}$	$\frac{1}{s-a}$
3.	$t^n, \quad n = 1, 2, 3, \ldots$	$\frac{n!}{s^{n+1}}$
4.	$t^p, p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5.	$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$
6.	$t^{n-\frac{1}{2}}, \quad n = 1, 2, 3, \ldots$	$\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7.	$\sin(at)$	$\frac{a}{s^2 + a^2}$
8.	$\cos(at)$	$\frac{s}{s^2 + a^2}$
9.	$t\sin(at)$	$\frac{2as}{(s^2 + a^2)^2}$
10.	$t\cos(at)$	$\frac{s^2 - a^2}{(s^2 + a^2)^2}$
11.	$\sin(at) - at\cos(at)$	$\frac{2a^3}{(s^2 + a^2)^2}$
12.	$\sin(at) + at\cos(at)$	$\frac{2as^2}{(s^2 + a^2)^2}$
13.	$\cos(at) - at\sin(at)$	$\frac{s(s^2 - a^2)}{(s^2 + a^2)^2}$
14.	$\cos(at) + at\sin(at)$	$\frac{s(s^2 + 3a^2)}{(s^2 + a^2)^2}$

15.	$\sin(at+b)$	$\dfrac{s\sin(b)+a\cos(b)}{s^2+a^2}$
16.	$\cos(at+b)$	$\dfrac{s\cos(b)-a\sin(b)}{s^2+a^2}$
17.	$\sinh(at)$	$\dfrac{a}{s^2-a^2}$
18.	$\cosh(at)$	$\dfrac{s}{s^2-a^2}$
19.	$\mathbf{e}^{at}\sin(bt)$	$\dfrac{b}{(s-a)^2+b^2}$
20.	$\mathbf{e}^{at}\cos(bt)$	$\dfrac{s-a}{(s-a)^2+b^2}$
21.	$\mathbf{e}^{at}\sinh(bt)$	$\dfrac{b}{(s-a)^2-b^2}$
22.	$\mathbf{e}^{at}\cosh(bt)$	$\dfrac{s-a}{(s-a)^2-b^2}$
23.	$t^n\mathbf{e}^{at}, \quad n=1,2,3,\ldots$	$\dfrac{n!}{(s-a)^{n+1}}$
24.	$f(ct)$	$\dfrac{1}{c}F\left(\dfrac{s}{c}\right)$
25.	$u_c(t)=u(t-c)$ **Heaviside Function**	$\dfrac{\mathbf{e}^{-cs}}{s}$
26.	$\delta(t-c)$ **Dirac Delta Function**	$\mathbf{e}^{-cs}$
27.	$u_c(t)f(t-c)$	$\mathbf{e}^{-cs}F(s)$
28.	$u_c(t)g(t)$	$\mathbf{e}^{-cs}\mathcal{L}\{g(t+c)\}$
29.	$\mathbf{e}^{ct}f(t)$	$F(s-c)$
30.	$t^nf(t), \quad n=1,2,3,\ldots$	$(-1)^nF^{(n)}(s)$
31.	$\dfrac{1}{t}f(t)$	$\int_s^\infty F(u)\,du$
32.	$\int_0^t f(v)\,dv$	$\dfrac{F(s)}{s}$

33.	$\int_0^t f(t-\tau)\,g(\tau)\,d\tau$	$F(s)\,G(s)$
34.	$f(t+T)=f(t)$	$\dfrac{\int_0^T \mathbf{e}^{-st} f(t)\,dt}{1-\mathbf{e}^{-sT}}$
35.	$f'(t)$	$sF(s)-f(0)$
36.	$f''(t)$	$s^2F(s)-sf(0)-f'(0)$
37.	$f^{(n)}(t)$	$s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$

Table Notes

1. This list is not a complete listing of Laplace transforms and only contains some of the more commonly used Laplace transforms and formulas.

2. Recall the definition of hyperbolic functions.

$$\cosh(t)=\frac{\mathbf{e}^t+\mathbf{e}^{-t}}{2} \qquad \sinh(t)=\frac{\mathbf{e}^t-\mathbf{e}^{-t}}{2}$$

3. Be careful when using “normal” trig function vs. hyperbolic functions. The only difference in the formulas is the “$+a^2$” for the “normal” trig functions becomes a “$-a^2$” for the hyperbolic functions!

4. Formula #4 uses the Gamma function which is defined as

$$\Gamma(t)=\int_0^\infty \mathbf{e}^{-x}x^{t-1}\,dx$$

   If n is a positive integer then,

$$\Gamma(n+1)=n!$$

   The Gamma function is an extension of the normal factorial function. Here are a couple of quick facts for the Gamma function